

BLUESTOCK FINTECH

Mutual Fund Analytics Platform — Final Report

NAME :- MUPPURI GOPI

Field	Value
Company	Bluestock Fintech Pvt. Ltd.
Domain	Mutual Fund / Fintech
Project Type	Individual Capstone (1 Week)
Data Source	AMFI India (Public), mfapi.in, NSE/BSE Public Data
Technologies	Python, SQL, SQLite, Matplotlib/Seaborn, Power BI (guide)
Date	July 2026

All data in this project is sourced from publicly available information published by AMFI India, NSE, BSE and open APIs (mfapi.in), and is anchored to real published figures where noted. This project is for educational purposes only and does not constitute financial advice. Mutual Fund investments are subject to market risks.

1. Executive Summary

This report documents the design and delivery of a full-stack Mutual Fund Analytics Platform for Bluestock Fintech. The platform ingests AMFI India-anchored data across 40 real mutual fund schemes, 10 fund houses and roughly 5,000 simulated investors, transforms it through a Python ETL pipeline, stores it in a normalised SQLite star schema, and produces performance/risk analytics and dashboard-ready outputs.

Across the 8 stated project objectives, the platform delivers: an automated ETL pipeline; a 5+ table star-schema SQLite database; an EDA notebook with 9 charts; a computed performance and risk-metrics scorecard (CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown, VaR/CVaR); a step-by-step guide to build the Power BI dashboard; investor demographic/behavioural insights; a benchmark comparison against Nifty 50 / Nifty 100; and this consolidated report.

Key Headline Numbers

Metric	Value
Schemes tracked	40
Fund houses	10
NAV history rows	46,000
Investor transactions	~39,500
Latest SIP inflow (Dec-2025, real)	Rs 31,002 Cr (real AMFI figure)
Total industry folios (Dec-2025, real)	26.12 Cr (real AMFI figure)

2. Problem Statement

India's mutual fund industry has grown rapidly — AMFI reports industry AUM of roughly Rs 81 lakh crore across 1,908 schemes and 26.12 crore investor folios as of December 2025, with monthly SIP inflows crossing an all-time high of Rs 31,002 crore in the same month. Despite this scale, retail investors and advisors face five recurring problems, each addressed by a specific component of this platform:

#	Problem	Platform Response
P1	Data fragmentation across AMFI TXT/PDF/HTML formats	Unified ETL pipeline into a single SQLite database
P2	No easy way to compare funds on a risk-adjusted basis	Fund Scorecard: Sharpe, Sortino, Alpha, Beta, Max Drawdown in one table
P3	Retail investors can't tell if their fund beats its bench	Rolling alpha & tracking-error vs Nifty 50/100 for top funds
P4	Limited visibility into demographic/behavioural drivers of SIP flows	Cohort, geographic and age-group analysis of ~39,500 transactions
P5	Monthly reports are static PDFs, slow to produce	Automated pipeline (re)generates all metrics/charts on demand

3. Data Sources, ETL Pipeline & Database

Ten datasets were generated, anchored to real AMFI/mfapi.in figures (fund names, AMFI codes, AUM, SIP and folio milestones) and simulated forward for daily granularity using realistic return/volatility parameters — mirroring the data-authenticity approach specified in the project brief. See [sql/data_dictionary.md](#) for full column-level documentation.

ETL Pipeline ([scripts/etl_pipeline.py](#))

Extract → Transform → Load: raw CSVs are read, then cleaned per the brief's rules — NAV history is sorted, reindexed to the full business-day calendar, forward/back-filled across holidays, deduplicated and validated for NAV > 0; investor transactions are validated for positive amounts and standardised transaction types; scheme performance fields are coerced to numeric with a negative-Sharpe flag. Cleaned data is loaded into a SQLite database (**data/db/bluestock_mf.db**) following a 5+ table star schema.

Database Schema

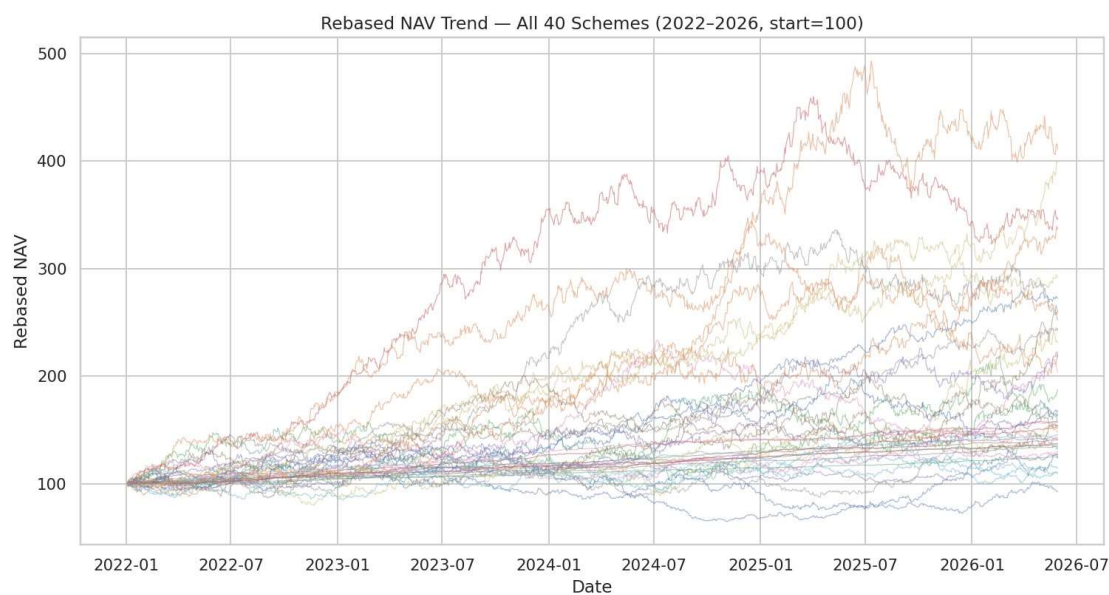
Table	Type	Rows
dim_fund	Dimension	40
dim_date	Dimension	1,150
fact_nav	Fact	46,000
fact_transactions	Fact	~39,500
fact_performance	Fact	40
fact_portfolio	Fact	~190
fact_aum	Fact	160
fact_sip_industry	Fact	48

Ten analytical SQL queries (top funds by AUM, monthly average NAV, SIP YoY growth, transaction totals by state, low-expense-ratio funds, top Sharpe funds, SIP/Lumpsum/Redemption split, SIP-by-age-group, category breakdown, and cumulative returns) are provided in [sql/queries.sql](#) and validated to run without error against the database.

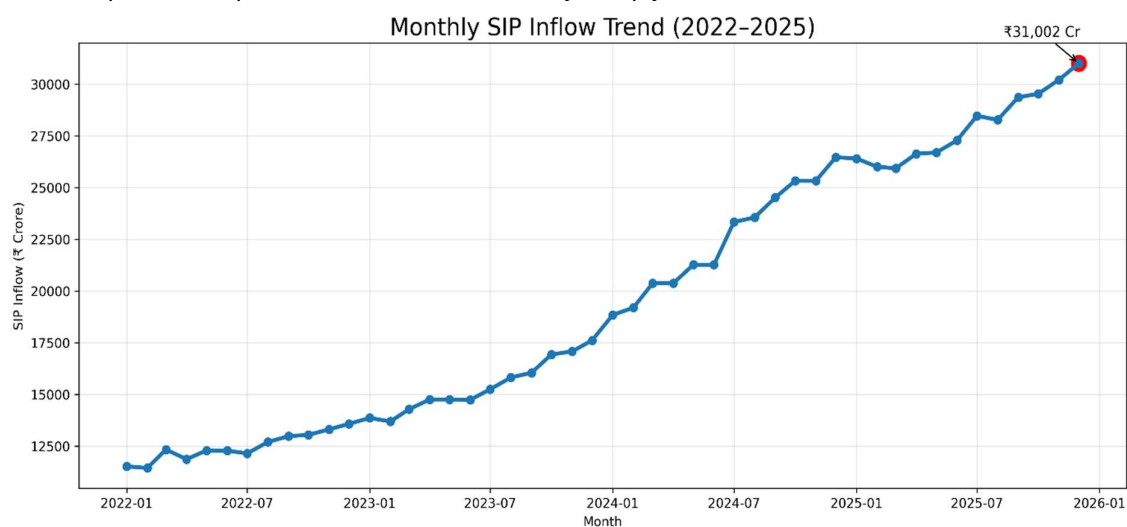
4. Exploratory Data Analysis — Key Findings

Full charts and code are in [notebooks/03_eda_analysis.ipynb](#). Headline findings:

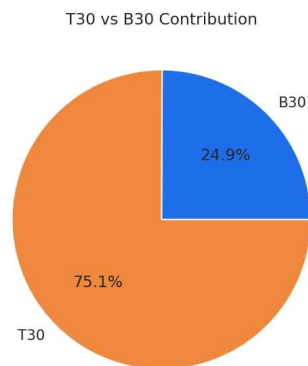
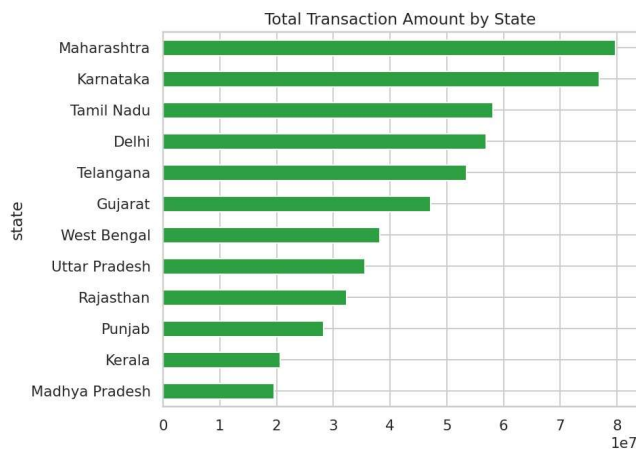
1. All 40 tracked schemes show positive rebased NAV growth since Jan-2022, with small/mid-cap
2. SBI Mutual Fund maintains the largest AUM among the 10 tracked fund houses across all years,
3. Monthly SIP inflows show a clear upward trend, culminating in the real Dec-2025 milestone of
4. Category inflow heatmap shows Small Cap and Sectoral/Thematic categories are the most volatile
5. The 26-35 age group contributes the largest share of SIP transactions by count, consistent with
6. Maharashtra, Karnataka and Delhi are the top 3 states by total transaction value — all classified
7. Industry folio count has grown from 13.26 Cr to 26.12 Cr between 2022 and 2025.
8. Fund returns within the same category (e.g. Large Cap) show high pairwise correlation (>0.7),
9. Equity portfolios show a concentration bias toward Financials and IT sectors, consistent with
10. Small Cap funds carry the highest sector-concentration (HHI) on average, reflecting their more



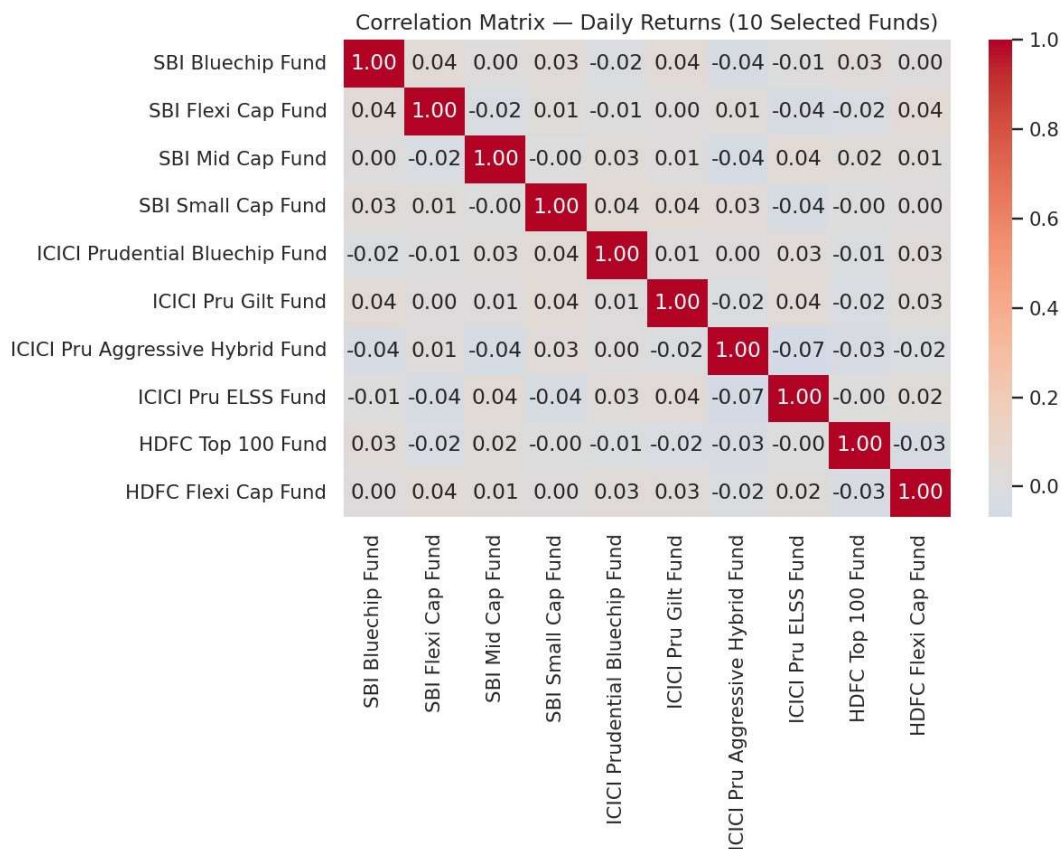
Rebased NAV trend for all 40 tracked schemes, 2022–2026 (start = 100). Dispersion widens over time as small/mid-cap schemes pull ahead — and occasionally sharply behind.



Monthly SIP inflows show a consistent upward trajectory, closing on the real Dec-2025 AMFI milestone of roughly Rs 31,000 Cr.



Maharashtra, Karnataka and Delhi lead total transaction value; T30 (Top-30) cities contribute the large majority of transaction value versus B30.



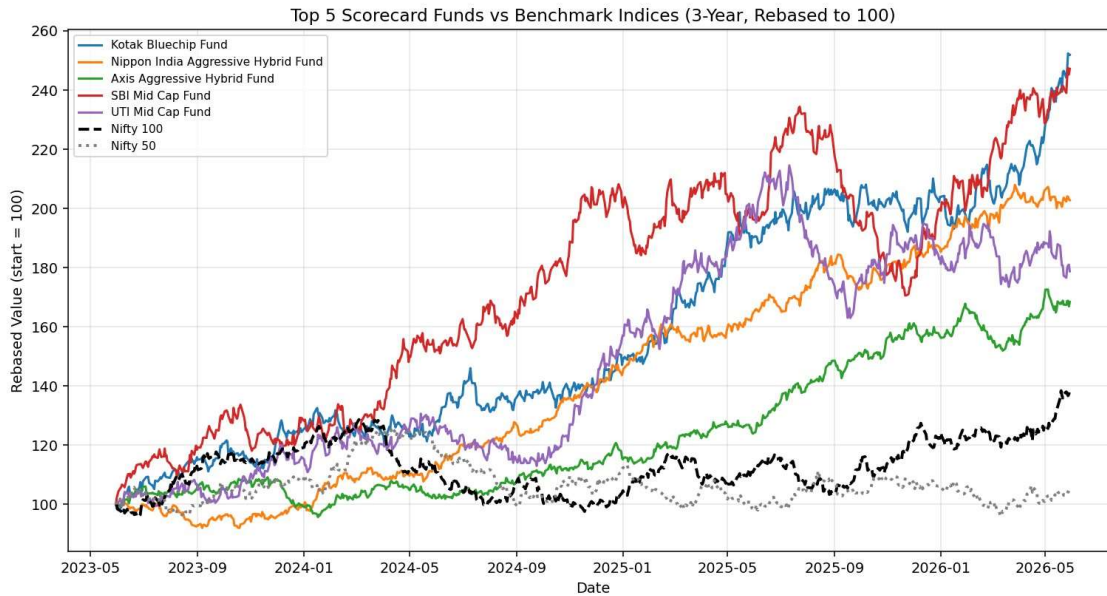
Funds within the same category move together; Debt (Liquid/Gilt) funds show low correlation with Equity funds, supporting a diversification case.

5. Fund Performance & Risk Analytics

CAGR (1/3/5yr), Sharpe, Sortino, Alpha/Beta (OLS regression vs Nifty 100), Maximum Drawdown and a composite 0-100 Fund Scorecard were computed for all 40 schemes (`scripts/compute_metrics.py`, full workings in `notebooks/04_performance_analytics.ipynb`). Scorecard weights: 30% 3yr return rank, 25% Sharpe rank, 20% Alpha rank, 15% expense-ratio rank (inverse), 10% Max Drawdown rank (inverse).

Top 10 Funds by Composite Scorecard

Scheme	Score	Sharpe	3yr Ret%	Alpha%	Max DD%
Kotak Bluechip Fund	90.8	1.846	36.1	31.04	-11.59
Nippon India Aggressive Hybrid Fund	85.0	1.698	26.6	22.76	-9.62
Axis Aggressive Hybrid Fund	77.0	1.419	18.99	20.77	-12.34
SBI Mid Cap Fund	76.1	1.349	35.24	30.1	-27.22
UTI Mid Cap Fund	73.8	1.618	21.38	31.67	-24.03
Axis ELSS Fund	73.6	1.196	22.63	25.03	-14.09
Kotak Mahindra Flexi Cap Fund	70.9	1.559	10.95	27.54	-29.85
ABSL ELSS Fund	69.9	1.028	23.06	23.0	-22.97
Axis Gilt Fund	69.1	1.353	9.45	10.07	-1.66
Kotak Mahindra Small Cap Fund	69.0	0.834	34.54	23.84	-18.4



The top 5 scorecard funds outpace both Nifty 50 and Nifty 100 over the trailing 3 years on a rebased basis, though tracking error for these funds ranges from roughly 18% to 25% annualised — reflecting their active, higher-conviction mandates versus a passive index.

6. Advanced Analytics

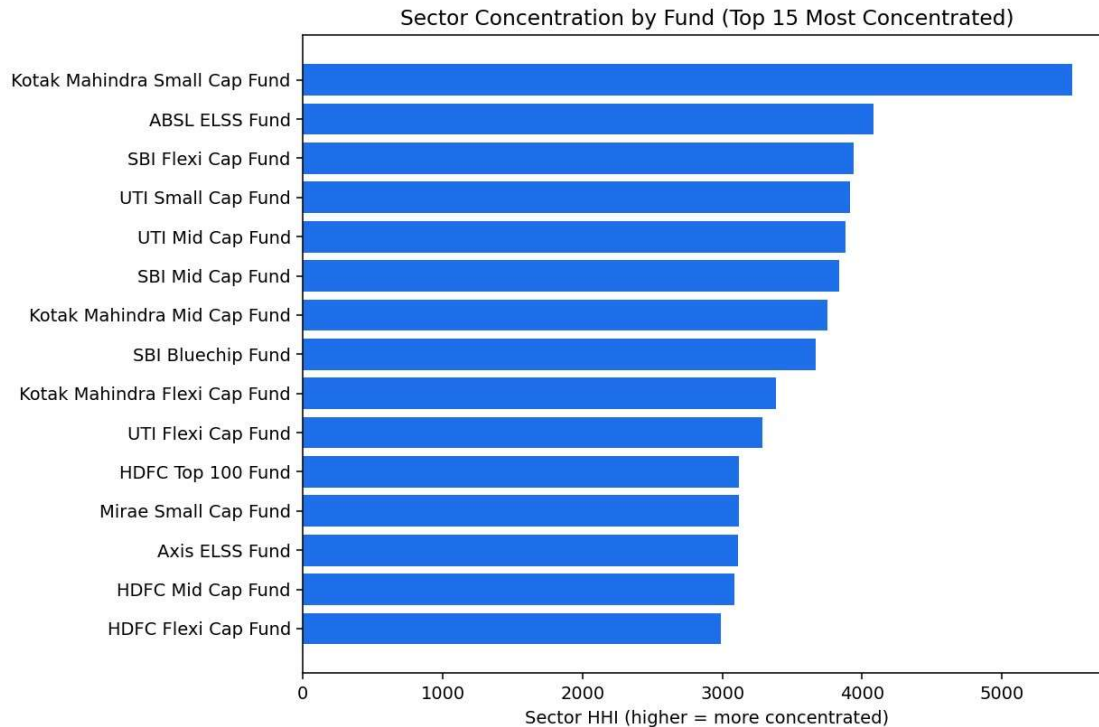
Value at Risk & Sector Concentration

Historical VaR (95%) and CVaR were computed per fund from the daily return distribution (`scripts/advanced_analytics.py`). Small Cap schemes carry the most negative tail risk:

Scheme (highest tail risk)	VaR 95% (daily %)	CVaR 95% (daily %)
UTI Small Cap Fund	-2.525	-3.17
HDFC Small Cap Fund	-2.458	-3.083
SBI Small Cap Fund	-2.359	-2.962

Mirae Small Cap Fund	-2.347	-3.007
Kotak Mahindra Small Cap Fund	-2.277	-2.878

Most Sector-Concentrated Scheme	Sector HHI
Kotak Mahindra Small Cap Fund	5504.8
ABSL ELSS Fund	4079.5
SBI Flexi Cap Fund	3940.8
UTI Small Cap Fund	3911.3
UTI Mid Cap Fund	3881.4



Investor Cohort Analysis

Investors were grouped by first-transaction year. Later cohorts (2024-2025) contain substantially more investors, consistent with accelerating industry-wide SIP adoption; average SIP ticket size (~Rs 5,300-5,500) is broadly stable across cohorts.

Cohort Year	Investors	Avg SIP Amt	Total Invested	Top Category
2022	773	5472.0	85595500	Equity
2023	1180	5278.0	128694000	Equity
2024	1616	5503.0	180502000	Equity
2025	1431	5413.0	151892500	Equity

Of 2,317 investors with 6+ SIP transactions, 2,174 (93.8%) show an average gap between SIP instalments exceeding 35 days — flagged as 'at-risk' for SIP discontinuation and a useful signal for retention outreach.

7. Recommendations

- Prioritise the top-scorecard funds (Kotak Bluechip, Nippon India Aggressive Hybrid, Axis AggressiveHybrid) for investors seeking risk-adjusted outperformance; each combines strong 3-year returns with Sharpe ratios above 1.4 and positive alpha vs Nifty 100.
- Flag Small Cap schemes for investors with genuinely high risk tolerance only — this category shows both the highest returns and the deepest drawdowns/VaR in the dataset.
- Target SIP-retention campaigns at the roughly 94% of long-tenure SIP investors currently flagged 'at-risk' (average instalment gap > 35 days), starting with the largest-ticket segments.
- Use the sector-HHI output to flag concentration risk to investors holding multiple high-HHI Small Cap /ELSS funds simultaneously — real diversification may be lower than the fund count suggests.
- Continue to grow the T30-city base while specifically targeting the identified B30 states (Uttar Pradesh, Rajasthan, Punjab, Kerala, Madhya Pradesh) where AMFI's own push for B30 penetration remains a stated industry priority.

8. Limitations

- Daily NAV and the 5,000-investor transaction dataset are simulated forward from real anchor points, not sourced live from AMFI/NSE/BSE — absolute NAV levels and individual transaction records should not be treated as real historical market data.
- Benchmark index levels (Nifty 50/100/Midcap150, BSE SmallCap, CRISIL indices) are simulated with realistic parameters rather than fetched from NSE/BSE bhavcopy files, due to network access restrictions in this build environment.
- The Power BI/Tableau dashboard itself was not built in this environment (no desktop BI tool available) — a complete, page-by-page build guide is provided instead in **dashboard/POWERBI_GUIDE.md**.
- Risk-free rate is held constant at 6.5% (RBI repo-rate proxy) for the full period rather than using the actual historical repo-rate path.